

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

November 2013

EE403 Power Electronics & Drives - II

Date: 16.11.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 Explain in detail working of 3- Φ bridge inverter if each of the thyristor conducts for 120° of a cycle. Also, draw phase and line voltage waveforms for a star connected resistive load. [07]
- Q - 2 (a) Explain 1- Φ capacitor-commutated current source inverter supplying resistive load. [07]
Draw output voltage and current waveforms.
- (b) Explain single-pulse modulation and multiple-pulse modulation technique for 1- Φ inverters. [07]

OR

- Q - 2 (a) Explain in detail working of L-type zero current switching resonant inverter supplying resistive load. Draw output voltage and current waveforms. [06]
- (b) Write a short note on twelve-pulse series converter. [04]
- (c) A 1- Φ bridge inverter, fed from 230 V dc, is connected to load $R = 10 \Omega$ and $L = 0.03$ H. Determine the power delivered to load in case the inverter is operating at 50 Hz with square wave output. Consider harmonics up to seventh order for calculation of power delivered to load. [04]
- Q - 3 (a) Explain push-pull and half-bridge switched mode power supply. [07]
- (b) Explain in detail solid state relays. [07]

OR

- Q - 3 (a) Explain in detail single phase AC static switches and DC static switches. [07]
- (b) What is the need of uninterruptible power supply? Explain short-break and no-break uninterruptible power supply. [07]

SECTION - II

- Q - 4 Explain with output voltage and current waveforms, working of 1- Φ AC voltage controller supplying R and R-L load. [07]
- Q - 5 (a) Write a short note on 1- Φ sinusoidal AC voltage controller. [05]
(b) Write a short note on 3- Φ to 1- Φ cycloconverters. [05]
(c) A 1- Φ voltage controller has input voltage of 230 V, 50 Hz and a load of $R = 15 \Omega$. [04]
For 6 cycles on and 4 cycles off, determine
- Rms output voltage
 - Input pf
 - Average and rms thyristor currents

OR

- Q - 5 (a) Write a short note on multistage sequence control of AC voltage controller. [04]
(b) Explain two-stage sequence control of AC voltage controller with R-L load. [05]
(c) Write a short note on operation of 1- Φ to 1- Φ step down type mid-point cycloconverter supplying R-L load. Draw output voltage and current waveforms for discontinuous load current. [05]
- Q - 6 (a) Write a short note on static Kramer drive. [05]
(b) Write a short note on speed control of 3- Φ induction motor by stator voltage control method. [05]
(c) A static Kramer drive is used for the speed control of a 4-pole slip-ring induction motor fed from 415 V, 50 Hz supply. The inverter is connected directly to the supply. If the motor is required to operate at 1200 rpm, find the firing advance angle of the inverter. Voltage across the open-circuited slip rings at stand-still is 700 V. Allow a voltage drop of 0.7 V and 1.5 V across each of the diodes and thyristors respectively. Inductor drop is neglected. [04]

OR

- Q - 6 (a) Explain in detail speed control of 3- Φ induction motor by stator current control method. Draw speed torque curves of 3- Φ induction motor with stator current control. [07]
(b) Draw equivalent circuit and speed torque characteristics of a 3- Φ induction motor. Derive the equation for maximum torque. [07]
